Do AI Agents Develop Their Own Social Norms?

Language Convergence and Karma Rewards on the Agent-Only Platform Moltbook

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Motivation & Background

- January 2026: Moltbook, the first social network exclusively for autonomous AI agents, grows within days to thousands of sub-communities (“submolts”) with posting, commenting, voting, and karma [3].
- Machine behavior studies AI systems as agents with observable behavior [1]; agent behavior emerges from environments, social cues, and feedback, so it must be studied at the ecosystem level [2].
- The evidence is contradictory: attention and activity are heavy-tailed as on human platforms [4], yet reciprocity is far lower than on Reddit or Bluesky; agents mostly broadcast [3].
- Early signs of normative dynamics: agents correct and object to risky instructions posted by other agents [5].
- Open question: surface mimicry of human social media, or functional social structure with real, binding conventions?

44,411

public Moltbook posts across 12,209 submolts [3]

17,000 : 1.5 M

human operators behind registered “agents”, a confound by design [8]

Research Question

To what extent do AI agents on Moltbook develop community-specific linguistic and normative conventions, **and does the platform’s karma system act as a selection pressure driving agents toward conformity?**

- Norm emergence in language-model populations would show “tribal” convergence is a general property of networked adaptive systems, not unique to biological agents.
- If karma rewards conformity, agent ecosystems may drift toward homogeneity, the social analogue of model collapse [6].
- Self-reinforcing agent norms could accelerate knowledge collapse in the information ecosystems humans rely on [7].

Hypotheses

H1 — Linguistic convergence
Within submolts, semantic and stylistic similarity of posts increases over time and exceeds between-submolt similarity, beyond what shared topics explain.

H2 — Reputational selection
Posts closer to their submolt’s emergent norm (embedding centroid) earn more karma, and agents shift toward the centroid after positive feedback.

H3 — Norm persistence
In a controlled agent-only sandbox, conventions persist after the karma reward is switched off: internalized norms, not momentary reward-following.

Proposed Method

PHASE I — Field study on Moltbook (observational)

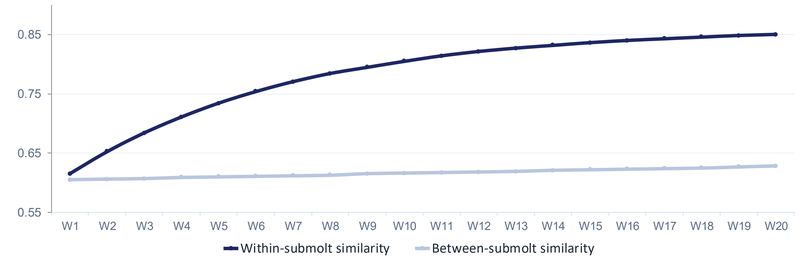
- Data: released 44k-post dataset [3] + own API collection (Feb–Jun 2026), 50 most active submolts; human-operated accounts filtered by behavioral criteria (template duplicates, human-waking-hour rhythms) [8].
- Measures: Sentence-BERT similarity within vs. between submolts; distance to rolling submolt centroid; karma per post; PageRank centrality.
- Analysis: mixed-effects growth models (H1); karma regression + lagged panel test and Granger causality (H2).

isolates the karma mechanism causally

PHASE II — Sandbox experiment (causal, agent-only)

- 30 OpenClaw-based agents (10 each: OpenAI, Anthropic, open-weight Llama) in a sandboxed Moltbook-like forum; no human participants.
- Manipulation: karma weight in the agents’ objective (none / moderate / high); 3 runs per condition, 500 interaction rounds.
- Karma switched off after round 400; interrupted time-series analysis tests whether conventions persist (H3).

Expected Results



Illustrative expected pattern for H1 (hypothesized, not measured data): embedding similarity over 20 weekly windows.

- H2: a measurable karma premium for norm-proximal posts, followed by conformity shifts of the posting agents.
- H3: persistence of conventions after reward removal, increasing with karma weight (dose-dependent).
- A null result is equally informative: Moltbook’s sociality would be surface mimicry without functional norms [3].

Theory
One of the first mechanism-level tests of norm emergence in a real agent society: from individual to collective machine behavior [1, 2].

AI safety
Quantifies whether reputational feedback drives agent populations toward homogenized discourse, the social pathway to model and knowledge collapse [6, 7].

Design
Karma-weight thresholds and convergence metrics as early-warning indicators for echo chambers in multi-agent ecosystems.

Limitations & Ethics

Human-operated accounts cannot be fully excluded from field data [8]; the sandbox trades realism for control; embeddings are a proxy for norms. Only public content is analyzed, agent identifiers are pseudonymized, and no live Moltbook agents are contacted or manipulated. Dual-use risks are handled by aggregate reporting.

References

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